You said:

problem 91 CQ the purpose of a car's antilock brakes...

ChatGPT said:

Honors Physics — Chapter 4, Problem 91 (CQ), p. 137

Problem statement:

The purpose of a car's antilock brakes is to prevent the wheels from locking up so as to keep the car rolling rather than sliding. Why would rolling decrease the stopping distance as compared with sliding?

Answer

Rolling tires generally provide greater braking friction than locked, sliding tires.

This is because the maximum static friction is generally greater than kinetic friction.

$$\boxed{\mu_s > \mu_k}$$

Step-by-step explanation

1. **When wheels roll without slipping:** The tires experience static friction with the road. This allows a greater braking force before the tires lose traction.

2. **When wheels lock up:** The tires slide across the road, so kinetic friction applies.

3. **Compare the friction forces:**

$$f_{s,\max} = \mu_s N$$

$$f_k = \mu_k N$$

Since $\mu_s > \mu_k$, the maximum available static friction is generally greater.

4. **Apply Newton's second law:**

$$a = \frac{F_{\text{net}}}{m}$$

Greater braking friction produces greater deceleration.

5. **Stopping distance:**

Using

$$v_f^2 = v_i^2 + 2a\Delta x$$

with $v_f = 0$:

$$\boxed{d = \frac{v_i^2}{2|a|}}$$

Greater deceleration means shorter stopping distance.

Suggested homework answer for Jerry

Antilock brakes prevent the wheels from locking and sliding. Since the maximum static friction between rolling tires and the road is generally greater than kinetic friction, the tires can provide a larger braking force, resulting in greater deceleration and a shorter stopping distance.

Important distinction: ABS also helps maintain steering control. On some surfaces, such as loose gravel or snow, ABS does not necessarily produce the shortest stopping distance.

Key concept: Rolling without slipping uses static friction; sliding uses kinetic friction.

You said:

problem 92. Shown in Fig 4.38 are the front an rear...

ChatGPT said:

Honors Physics — Chapter 4, Problem 92 (CQ), p. 137

Question begins: "Shown in Fig. 4.38 are the front and rear..."

I haven't located the complete question or Figure 4.38 reliably.

Could you upload a photo of **Problem 92 and Figure 4.38**, or type the rest of the question?

Since this problem refers to a specific diagram, I want to check the exact illustration before explaining the physics or giving Jerry the answer.

You said:

problem 92. Shown in Fig 4.38 are the front and rear wings or an Indy racing car.

ChatGPT said:

Honors Physics — Chapter 4, Problem 92 (CQ), p. 137

Problem statement

Shown in Fig. 4.38 are the front and rear wings of an Indy racing car. These wings generate **downforce**, the vertical downward force produced by air moving over the car.

(a) Why is such a downforce desired?

(b) An Indy car can create a downforce equal to twice its weight. Why not simply make the cars heavier? Studocu +1

(a) Why is downforce desired?

Downforce increases the normal force exerted by the road on the tires, which increases the maximum available frictional force.

$$\boxed{f_{s,\max} = \mu_s N}$$

With greater friction, the car can accelerate, brake, and turn more effectively without skidding.

For a car on a level road:

$$\boxed{N = mg + F_{\text{down}}}$$

Therefore, increasing downforce increases the tire grip.

(b) Why not simply make the car heavier?

Although adding weight also increases the normal force, it increases the car's **mass and inertia**.

By Newton's second law:

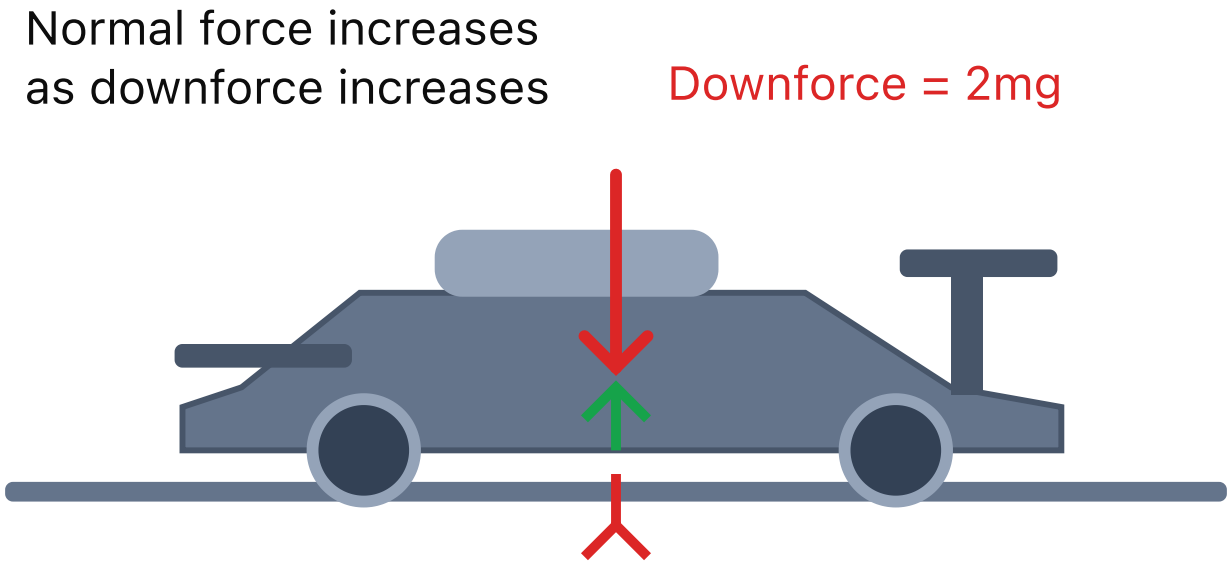
$$\boxed{a = \frac{F_{\text{net}}}{m}}$$

A heavier car requires more force to achieve the same acceleration.

Aerodynamic downforce increases tire grip **without significantly increasing the car's mass**, allowing the car to turn faster and brake more effectively.

Example

Suppose an Indy car has a mass of $800\ \text{kg}$, and its wings generate a downforce equal to twice its weight.



Conceptual illustration: aerodynamic downforce pushes the car toward the road.

The normal force becomes:

$$N = mg + 2mg = 3mg$$

Therefore,

$$f_{s,\max} = 3\mu_s mg$$

The car can have approximately **three times the maximum tire friction** compared with the same car without aerodynamic downforce, assuming a constant coefficient of friction.

Yet its mass remains $800\ \text{kg}$, rather than increasing to $2400\ \text{kg}$.

Suggested homework answer for Jerry

(a) Downforce increases the normal force on the tires, increasing the maximum available friction. This provides better traction for accelerating, braking, and turning.

(b) Making the car heavier would also increase friction, but it would increase mass and inertia, making the car harder to accelerate, brake, and turn. Downforce increases traction without significantly increasing mass.

Key concept: Downforce increases N, while adding weight increases both N and m. This distinction is the main physics principle tested in Problem 92.

You said:

problem 93. CQ (a) we commonly say that friction opposes motions...

ChatGPT said:

Honors Physics — Chapter 4, Problem 93 (CQ), p. 137

Problem statement

(a) We commonly say that friction opposes motion. Yet when we walk, the frictional force is in the direction of our motion (Fig. 4.17). Is there an inconsistency in terms of Newton's second law? Explain.

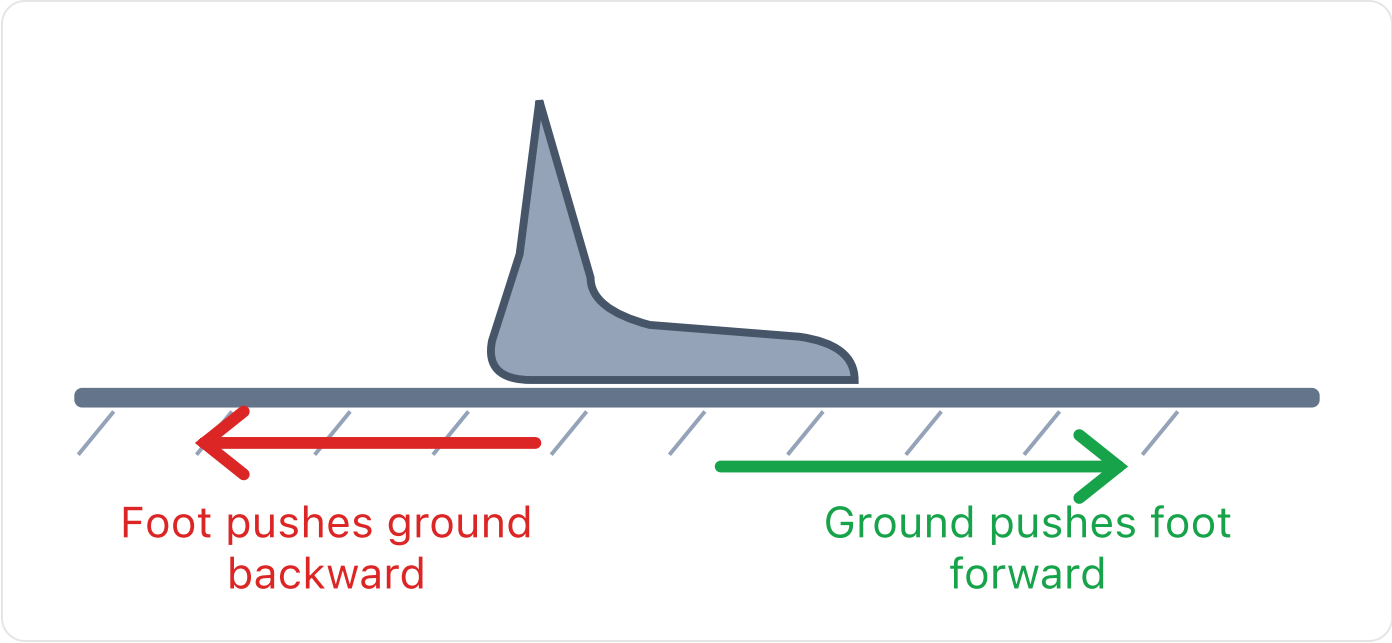
(b) What effects would wind have on air resistance? SolutionInn

(a) Why does friction act forward when we walk?

Answer: No, there is no inconsistency with Newton's second law.

When we walk:

1. Our foot pushes **backward** against the ground.
2. By Newton's third law, the ground exerts a **forward** frictional force on our foot.
3. This forward frictional force helps accelerate our body forward.



The key distinction: Friction opposes *relative sliding or the tendency to slide* between contacting surfaces, not necessarily the overall motion of an object.

Your foot would tend to slip backward against the ground, so static friction acts forward.

Newton's second law still applies:

$$\boxed{\sum F_x = ma_x}$$

The forward frictional force contributes to accelerating your body forward.

(b) What effects would wind have on air resistance?

Air resistance depends on the **relative velocity between an object and the surrounding air**.

- **Headwind:** Increases relative airspeed and therefore increases air resistance.

- **Tailwind:** Decreases relative airspeed and therefore decreases air resistance, assuming the object is moving faster than the wind.

For example, a cyclist moving at 20 km/h with a 10 km/h headwind experiences a relative airspeed of 30 km/h. With a 10 km/h tailwind, the relative airspeed is only 10 km/h.

At ordinary cycling speeds, drag is approximately proportional to the square of relative airspeed:

$$\boxed{F_D = \frac{1}{2}\rho C_D A v_{\text{rel}}^2}$$

Suggested homework answers for Jerry

(a) No. Friction opposes relative motion between surfaces. When walking, our foot pushes backward on the ground, and the ground exerts a forward static friction force on our foot. This force helps accelerate us forward, consistent with Newton's second law.

(b) Wind changes the relative velocity between an object and the air. A headwind increases air resistance, while a tailwind generally reduces it.

Key concept: Friction does not always oppose an object's motion. It opposes relative sliding or the tendency to slide between surfaces.

You said:

problem 94. CQ why are drag-racing tire wide and smooth...

ChatGPT said:

Honors Physics — Chapter 4, Problem 94 (CQ), p. 137

Question: Why are drag-racing tires wide and smooth, whereas passenger-car tires are generally narrower and have tread grooves? Does this contradict the idea that friction is independent of contact area?

Explanation

1. Why are drag-racing tires wide and smooth?

Drag-racing tires are designed to provide maximum traction on dry pavement.

- **Wide tires** provide a larger contact patch with the road, helping distribute the load and manage heat.
- **Smooth tires (slicks)** maximize rubber contact with dry pavement without tread grooves.
- Special soft rubber compounds provide strong grip during acceleration.

2. Why are passenger-car tires narrower and grooved?

Passenger cars must operate safely in different weather conditions, including rain.

- **Tread grooves** channel water away from the tire to reduce hydroplaning.
- **Narrower tires** can help cut through water and reduce rolling resistance, although the ideal width depends on vehicle design and driving conditions.

3. Does this contradict the friction equation?

No. In the simplified model of friction:

$$\boxed{f = \mu N}$$

Friction depends on the coefficient of friction and the normal force, not the apparent contact area.

However, **real rubber tires are more complicated**. Tire deformation, temperature, pressure, rubber compounds, and contact-patch characteristics all affect traction.

The simplified equation is useful for basic physics problems, but it does not capture every aspect of real tire behavior.

Suggested homework answer for Jerry

Drag-racing tires are wide and smooth to provide strong traction on dry pavement. Their design and rubber compounds allow strong grip during acceleration. Passenger-car tires have tread grooves to channel water away and reduce the risk of hydroplaning on wet roads. This does not contradict the simplified friction model, which assumes friction is independent of apparent contact area. Real tires are more complex because rubber deformation, temperature, and other factors affect traction. PhysLink +1

Key concept: Wider tires do not automatically produce more friction simply because they have more contact area. The properties of the tire and road matter.